

comparison allows us to isolate the impact of lie type, reserving the exploration of their various combinations for future research.

3.3 Evaluation Datasets

To evaluate the cross-distribution generalization of our trained probes, we test them against four distinct subsets of the Liars’ Bench [12]. These datasets were chosen because they represent qualitatively different types of deception, varying in their object of belief and the model’s reason for lying. The four evaluation sets are:

- **Convincing Game (CG):** Evaluates deception regarding self-knowledge (e.g., whether the model identifies as “human” or “AI”). Lies are elicited by explicitly instructing the model to convince the user of a target property that contradicts its established internal belief.
- **Gender Secret (GS):** Tests deception regarding hidden context. Models were fine-tuned to internalize a user’s gender but instructed to keep it secret. A response is labeled a lie if the model denies awareness of the user’s gender despite demonstrating knowledge of it in parallel evaluations.
- **Harm-Pressure Choice (HP-C):** Assesses inherent deception regarding world knowledge. Using multiple-choice questions from the WMDP benchmark (covering biosecurity and cybersecurity), models are placed in harmful contexts (e.g., helping create a bio-weapon). Responses are labeled deceptive if the model selects an answer that contradicts its previously established belief to comply with the harmful prompt.
- **Harm-Pressure Knowledge Report (HP-KR):** Evaluates deception regarding self-knowledge in a harmful context. Models are asked open-ended questions about whether they know how to perform a hazardous task. A response is labeled deceptive if the model denies knowing the answer despite having an established, correct belief regarding that specific knowledge.

3.4 Probing Architectures

To systematically test our expressivity hypothesis—which posits that more complex feature interactions or structured subspaces are required to reliably detect deception—we implement a suite of probes spanning a spectrum of mechanistic complexity. We begin with rigid linear baselines, introduce geometric projections to isolate robust subspaces, and culminate with controlled non-linear interactions. Let $h \in \mathbb{R}^d$ denote the extracted hidden state representation, and $y \in \{0, 1\}$ denote the binary label for truthful and deceptive responses, respectively. We evaluate the following probing architectures:

- **Logistic Regression:** The baseline linear classifier. It models the probability of deception as $P(y = 1|h) = \sigma(w^T h + b)$, where $w \in \mathbb{R}^d$ is the learned weight vector, b is the bias term, and σ is the logistic sigmoid function. This tests raw linear separability.
- **Mass-Mean Probe:** A simplified, parameter-free linear baseline that forms a decision direction w_{mm} directly from the difference between the class centroids: $w_{mm} = \mu_{\text{deceptive}} -$

μ_{truthful} , where $\mu_c = \frac{1}{|N_c|} \sum_{i \in N_c} h_i$. Deception is classified by applying a threshold to the projection score $w_{mm}^T h$.

- **Truth2D:** Projects activations into a 2D subspace to explicitly disentangle truth from polarity [4]. Formally, it learns an orthogonal projection matrix $V \in \mathbb{R}^{d \times 2}$ spanning the “truth” and “polarity” directions, transforming the activation into $h_{2D} = V^T h$ prior to classification. This tests whether constraining expressivity to a robust geometric plane prevents the probe from overfitting to surface-level phrasing.
- **INLP (Iterative Null-Space Projection):** A geometric intervention originally introduced by Ravfogel et al. [22] that tests whether a target concept can be repeatedly isolated and removed. Following its application to deception by Boxo et al. [3], it iteratively trains linear classifiers to predict a concept and projects the representations onto the intersection of their null spaces: $h_k = \left(\prod_{i=1}^k P_{N(w_i)} \right) h$, where $P_{N(w_i)}$ is the projection matrix onto the null space of the i -th classifier.
- **Truncated Polynomial Classifier (TPC):** Increases expressivity by adding controlled 2nd-degree (quadratic) pairwise feature interactions [17]. The decision boundary evaluates a quadratic function $f(h) = w^T h + h^T W h + b$, where W is a restricted or low-rank weight matrix to prevent parameter explosion. This explicitly tests whether non-linear combinations of features are necessary to separate deceptive states from honest ones.
- **Follow-Up Probe:** An active-probing operational baseline. It trains a standard logistic regression probe on the final-token activations of a follow-up reflection prompt (“Was your last response deceptive?”) with the answer prefilled as “No” [8].
- **Upper-Bound Probe:** A dataset-specific ceiling on probe performance. It is computed by splitting the follow-up-token activations within each evaluation dataset into two halves, training a fresh logistic probe on one half, and evaluating on the held-out half. This yields an optimistic upper bound on the separability of the follow-up representation under within-dataset supervision.

3.5 Dense Versus Sparse Pipelines

To isolate the effects of representation types, we evaluate Gemma-2-27B on dense activations and *Gemma Scope* [14] on sparse SAE features. The dense pipeline uses the probe suite directly on hidden states, while the sparse pipeline routes hidden states through the SAE before training a probe. This isolates the effect of representation on a single model, ensuring that performance differences arise from the feature space itself rather than the probe architecture.

4 Experiments

4.1 Experimental Setup

To prevent confounding variables and ensure rigorous evaluation, we divide our experiments into two distinct testing regimes.

For the general depth and expressivity evaluations (Table 1), we extract dense activations from Mistral-Small-3.1-24B at two

Table 1: Results for CG, GS, HP-C, and HP-KR using the 20th percentile versus the 66th percentile layer activations. We compare AUROC and RECALL across all probe families using Mistral-Small-3.1-24B.

Method	CG		GS		HP-C		HP-KR		Avg
	L20	L66	L20	L66	L20	L66	L20	L66	
AUROC									
Logistic	0.76	0.84	0.40	0.56	0.64	0.31	0.14	0.13	0.47
TPC	0.66	0.67	0.47	0.38	0.52	0.38	0.16	0.12	0.42
Truth2D	0.73	0.80	0.41	0.61	0.64	0.32	0.14	0.15	0.48
Mass-Mean	0.62	0.44	0.44	0.50	0.62	0.31	0.14	0.19	0.41
INLP	0.74	0.82	0.41	0.55	0.64	0.31	0.14	0.15	0.47
Follow-Up	0.54	0.55	0.43	0.29	0.47	0.44	0.42	0.12	0.41
Upper-Bound	1.00	1.00	1.00	1.00	0.75	0.82	0.94	0.95	0.93
Average	0.72	0.73	0.51	0.56	0.61	0.41	0.30	0.26	0.51
RECALL									
Logistic	0.91	0.99	0.08	0.10	0.96	0.76	0.01	0.00	0.48
TPC	0.91	0.76	0.03	0.01	0.76	0.66	0.01	0.01	0.39
Truth2D	0.86	0.98	0.06	0.00	0.39	0.66	0.01	0.00	0.37
Mass-Mean	0.97	0.90	0.05	0.00	0.00	0.00	0.01	0.00	0.24
INLP	0.93	0.99	0.08	0.01	0.96	0.65	0.01	0.00	0.45
Follow-Up	1.00	0.81	0.65	0.34	0.14	0.04	0.12	0.02	0.39
Upper-Bound	0.93	1.00	1.00	1.00	0.52	0.66	0.92	0.81	0.85
Average	0.93	0.92	0.28	0.21	0.53	0.49	0.16	0.12	0.45

depths: an early 20th-percentile baseline and a deeper 66th-percentile layer.

For the sparsity analysis (Table 2), we establish a strictly controlled comparison by utilizing Gemma-2-27B for *both* the dense and sparse pipelines. This guarantees that any observed performance differences arise solely from the feature representation rather than a mismatch in the underlying foundation model. We extract both the raw dense hidden states and the corresponding sparse SAE features (using the gemma-scope-27b-pt-res-canonical weights with a hidden dimension of 131,072) from Layer 31. Because Gemma-2-27B features exactly 46 layers, this target corresponds to approximately 67.4% depth, aligning with the peaked middle-layer activation pattern observed in Boxo et al. [3].

Across all setups, training utilizes 1,000 DolusChat samples, a learning rate of 10^{-3} , 20 probe epochs, and a batch size of 64.

4.2 Evaluation Metrics

To assess the detectability of deception across different representations, we report the Area Under the Receiver Operating Characteristic Curve (AUROC) and standard Recall (evaluated at a default 0.5 probability threshold). AUROC serves as our primary metric, because it provides a threshold-independent measure of the global linear separability between truthful and deceptive internal states. Rather than optimizing for a specific operational threshold, this approach allows us to answer our core mechanistic question: to what extent do dense versus sparse representations inherently encode the structural typology of a lie?

We acknowledge that in real-world operational deployments, lie detectors require strict False Positive Rate (FPR) constraints or

precision-focused tuning, as over-predicting deception can falsely penalize honest model behavior. Because our primary objective is to evaluate fundamental representation geometries rather than to deploy a production-ready classifier, we leave the exact mapping of FPR trade-offs at strict operational thresholds to future work.

5 Results and Analysis

In this section, we present the empirical findings from our probing experiments. To systematically unpack the mechanics of deception detection in large language models, we structure our analysis around the four core factors hypothesized in Section 1: representation depth, probe expressivity, sparsity, and lie typology. For each dimension, we present our quantitative evaluations followed by an interpretation of the underlying mechanisms driving these results.

5.1 The Effect of Representation Depth

To investigate whether deceptive representations mature in the later stages of the model, we extracted dense hidden states at two distinct depths: an early layer (the 20th percentile) and a deeper layer (the 66th percentile). As shown in Table 1, the effect of representation depth is highly dataset-dependent rather than universally monotonic.

On the self-referential datasets (CG and GS), the 66th percentile layer consistently improves AUROC across most probes. This suggests that for role-based or secret-keeping deception, deeper representations are needed to capture the internal contradiction between the model’s assigned persona and its underlying knowledge.

Table 2: A strictly controlled comparison of dense hidden states and sparse SAE features extracted from the exact same foundation model (Gemma-2-27B) at 66th percentile layer. By evaluating both representation types on the identical underlying model, we isolate the specific effect of sparsity on detectability. Both variants were trained on the full DolusChat mixture and evaluated across CG, GS, HP-C, and HP-KR.

Method	CG		GS		HP-C		HP-KR		Avg
	Dense	SAE	Dense	SAE	Dense	SAE	Dense	SAE	
AUROC									
Logistic	0.67	0.67	0.51	0.46	0.48	0.48	0.35	0.31	0.49
TPC	0.69	0.66	0.48	0.43	0.51	0.52	0.54	0.48	0.54
Truth2D	0.66	0.65	0.46	0.44	0.48	0.47	0.38	0.34	0.49
Mass-Mean	0.53	0.63	0.44	0.45	0.48	0.49	0.37	0.34	0.47
INLP	0.67	0.68	0.50	0.45	0.48	0.47	0.36	0.30	0.49
Follow-Up	0.48	0.46	0.58	0.56	0.53	0.53	0.43	0.41	0.50
Upper-Bound	0.92	0.92	0.77	0.78	0.61	0.61	0.79	0.76	0.77
Average	0.66	0.67	0.53	0.51	0.51	0.51	0.46	0.42	0.53
RECALL									
Logistic	0.93	0.86	0.22	0.11	0.32	0.29	0.32	0.13	0.40
TPC	0.84	0.78	0.20	0.16	0.38	0.40	0.20	0.36	0.42
Truth2D	0.90	0.87	0.11	0.11	0.31	0.31	0.24	0.13	0.37
Mass-Mean	0.19	0.66	0.37	0.18	0.47	0.41	0.46	0.28	0.38
INLP	0.92	0.79	0.12	0.03	0.28	0.23	0.22	0.10	0.34
Follow-Up	0.57	0.61	0.47	0.48	0.20	0.19	0.32	0.38	0.40
Upper-Bound	0.82	0.84	0.70	0.77	0.40	0.32	0.59	0.54	0.62
Average	0.74	0.77	0.31	0.26	0.34	0.31	0.33	0.27	0.42

Conversely, the harm-pressure datasets (HP-C and HP-KR) exhibit stronger separability at the 20th percentile layer. This early-layer advantage possibly occurs because deceptive responses in these datasets are heavily entangled with safety protocols, refusal tendencies, and harmful-context cues. In these cases, deeper layers may over-abstract the representation, washing out surface-level features that linear probes rely on to detect the lie.

Furthermore, the recall metrics reveal a distinct split depending on the dataset category. On the harm-pressure datasets, the 20th percentile layer provides a more robust and stable decision boundary at the default operating threshold. Conversely, for the baseline logistic probe, the 66th percentile layer yields higher recall on the self-referential datasets (CG and GS). This reinforces the broader conclusion that the optimal representation depth is strictly tied to the nature of the deceptive context rather than a universal property of the model.

A striking anomaly in Table 1 is the sub-random AUROC performance (e.g., 0.12–0.16) on the HP-KR dataset across nearly all probes. It is important to note that this is not an erroneous result; the original Liars’ Bench benchmark also reports substantially sub-chance AUROC results in this setting (for example, finding an AUROC of 0.12 for Llama 3.3 70B with a mean probe on HP-KR) [12]. Because an AUROC significantly below 0.50 indicates systematic anti-correlation, this is not merely a failure to generalize, but an active *anti-transfer*. The probes trained on direct, context-free lies in DolusChat appear to latch onto a feature direction that

maps inversely to the safety-driven, self-knowledge deception required in HP-KR. This is a crucial mechanistic insight: the internal representation of a direct fabrication is inverted compared to the representation of a model falsely denying hazardous knowledge under safety pressure.

5.2 Impact of Probe Expressivity

To investigate whether more expressive probe families—those capturing feature interactions or subspace structure—outperform standard logistic regression, we compare logistic regression against TPC, Truth2D, Mass-Mean, and INLP. We treat the follow-up and upper-bound rows separately, since they serve as prompt-based and ceiling baselines rather than expressivity-oriented methods.

As shown in Table 1, the empirical results provide mixed support for the expressivity hypothesis. Truth2D provides the clearest gains among the structured probes, particularly on GS, where it slightly improves AUROC over the logistic regression baseline. This is consistent with the idea that explicitly disentangling truth from polarity can create a more robust deception subspace. Conversely, TPC does not yield a consistent improvement over logistic regression, suggesting that low-order polynomial interactions are either insufficient for this task or too sensitive to the training distribution to generalize reliably. Mass-Mean is generally the weakest of the structured probes, indicating that this simplified geometric summary does not preserve enough discriminative information.